

What makes R better?

In a word - customisation. There are other languages and applications that analyse data, but none offer the flexibility or range in packages that R does. Applications like MATLAB, S-PLUS, Stata, SPSS and SAS hide their command prompts behind fancy GUIs. These products are owned and developed by corporations like IBM that design and copyright the source code behind their GUIs.

This admittedly makes most data modelling easier as the learning curve is quicker. The user has only to learn which button to click on to get desired results along with how to interpret results. However, this only works as long as the user does not need any additional customisation requiring a change in the source code, which would be impossible to access as this is proprietary software. In this respect R has a huge advantage as it is open source and the source code is freely available.

Languages like Python, Java and Julia can also model and visualise data, but they have not been around as long as R has and so lack the large library of statistical and graphics packages that R offers. For example, Python still lacks the ability to run multiple regression scripts. It will take a few years for its library to catch up with that of R.

How do you get R?

The base R programming language is available as a free download from <http://www.r-project.org>.

R can be installed on Windows, Mac, Unix, Linux systems and even on the PlayStation 3 (though this is only for serious geeks who don't mind missing out on gaming time). After downloading R you can also choose to download R Studio, which has a better interface, with multiple screens to work between. You can write scripts in a separate text box, check a list of defined variables, past commands, look at graphs and help files, and code from the command prompt all at the same time.

To download R, go to <http://www.r-project.org> > Click on 'download R' > click on a mirror link to download from the server closest to where you live. For India, this will be located at IIT Chennai (still referred to as Madras on the R site).

To download R Studio, go to <http://www.rstudio.com/products/rstudio/download/> > click on the latest installer for your system. Remember that you need to have already downloaded R first.

The R download, called the base system, contains the R programming language, a compiler, debugger, useful datasets, and a number of common